Response to the Reviewers

The waste collection routing problem with time-varying stochastic demand: A machine learning based approach
(Ms. Ref. No.: CAIE-D-26-05414)

We would like to thank the Associate Editor and the Reviewers for allowing us to submit an improved version of our paper.

We have carefully considered all of the suggestions and addressed all concerns. In the following, we provide our response to the two Reviewers (whose comments appear in **boldface**). The modified text in the revised manuscript appears in blue font.

We hope this new version will be appreciated.

Reviewer #1

The paper addresses a relevant waste collection routing problem by combining intraday stochastic demand modelling with machine learning for policy selection. The approach is promising, and the paper merits publication after the following revisions.

1. **Please proofread the manuscript carefully. For example, replace "Firt-stage" with "First-stage," "A such feature" with "Such a feature," and "the trained model predict" with "the trained model predicts." Check subject–verb agreement, articles, and terminology throughout.**

   Francesca

2. **Clarify how constraint (19) ensures mandatory skipping after capacity saturation. Its upper bound permits the skipping variable to remain zero even when the corresponding arc is selected and saturation occurs. Explain whether the objective function guarantees the intended behaviour under stated cost assumptions, or introduce the necessary complementary constraint.**

   Francesca We thank the Reviewer for the comment and we agree for the need to better clarify...

3. **Provide valid numerical bounds for M and Qmax, and clarify the saturation convention when cumulative load equals vehicle capacity in constraints (17)–(18).**

   Giuseppe

4. **Reconcile the reported average VSS of 21.33% with Table 5. Weighting the displayed group averages by their respective numbers of solved instances gives approximately 27.02%. Please verify the calculation and update the discussion and conclusions accordingly.**

   Giuseppe

5. **Moderate claims concerning generalisation and practical scalability. The experiments involve five to eight collection points, five demand scenarios, and 63 test instances for the machine learning assessment. These limitations should be acknowledged explicitly.**

   Francesca

6. **Strengthen the discussion of stochastic routing and recourse approaches by considering the following relevant references. Explain how the manuscript's intraday demand modelling and policy selection distinguish its contribution from these studies.**
   *** Jammeli, H., Argoubi, M., & Masri, H. (2021). "A Bi-objective stochastic programming model for the household waste collection and transportation problem: case of the city of Sousse." Operational Research, 21, 1613–1639. https://doi.org/10.1007/s12351-019- 00538-5**
   *** Masri, H., Ben Abdelaziz, F., & Alaya, H. (2016). "A Recourse Stochastic Goal Programming Approach for the Multi-objective Stochastic Vehicle**

Routing Problem." Journal of Multi-Criteria Decision Analysis, 23(1–2), 3–14. https://doi.org/10.1002/mcda.1563

Francesca

Reviewer #2

The topic of the paper is interesting and the overall idea can be methodologically useful for other type of problems as well. But I have some reservations about the paper.

1. **First of all, the main distinction of the paper is stated as the intra-day nature of the stochasticity they model. However, this distinction should be explain in a much sharper way. The time-dependent nature mentioned from the literature which is around the concept stochasticity across days, rather than intra-day, methodologically could be similar to represent. What is the mathematical difference and therefore the challenge associated with intra-day with respect to those that assume stochastocitiy across days? This needs to be clarified from a mathematical formulation point of view in order to justify the contribution claims. Related to the contributions, Table 1 is a little misleading I think. For example, Markov et al. also consider time dependency but not visible from this Table. May be better to add another column to distinguish this aspect across papers?**

   Giuseppe (e, successivamente, Francesca)

2. **Another important comment I have is in relation to the presented results in Section 5.4. What is the understanding with respect to the problem characteristics? Can you interpret the learning with respect to the policy selection? Which characteristics go with what type of policy? I thought this was the goal and that there would be interpretations around this. To me, without an elaborate discussion of the mapping between the policies and the problem characteristics, the mentioned goals are not complete. If we want to move those ideas to operations, this interpretation should be there.**

   Alessio

3. **In section 4.2 the authors list the features they considered. For the set of the features the authors used here, is there any evidence that this is indeed performing better than other potential feature sets? Was there any analysis on this? Furthermore, the definition of certain features are also based on certain insights I assume, for example the percentage of collection points etc. have specific descriptions. Would be good to elaborate on this further.**

   Alessio

4. **Otherwise, the paper is quite well written and easy to follow. I read it with great interest. One small thing i noticed, at the top of page 4, 2nd line it says: "The latter also One small thing i noticed, at the top of page 4, 2nd line it says: "The latter also consider...", it should be "The latter also considers..."**

   Francesca

Reviewer #3

Indicati esplicitamente il punto assegnato ad Alessio e il punto assegnato a Giuseppe. Il resto è assegnato a Simona, per come concordato.

I thank the authors for the opportunity to review this manuscript. The paper addresses a relevant waste collection routing problem, and the interaction between collection time, uncertain waste accumulation and vehicle capacity is potentially valuable. However, the manuscript does not yet sufficiently establish the significance of its contribution or provide convincing computational validation.
My principal concerns involve the positioning against existing models and algorithmselection approaches, clarification of the stochastic information structure, and experiments restricted to very small instances with limited scenario and generalisation testing.
Addressing these concerns appears to require substantial further investigation and an expanded computational study. I therefore recommend rejection in its present form. I hope the detailed comments below will help the authors strengthen the work.

1. **Abstract**

   - **The abstract primarily describes the problem. Please identify the research gap and state the specific contributions relative to existing stochastic and time-dependent waste collection routing studies.**
   - **The phrase "traditional methodology employing scenarios" is vague. Please identify the formulation and solution approach more precisely.**
   - **Please clarify the role of machine learning: what the candidate policies represent, what the learning model predicts and why policy selection is useful.**
   - **Although computational experiments are mentioned, their principal findings are not reported. Please include the main quantitative results and identify the comparison baseline.**
   - **Please specify that synthetic instances are used and indicate their scale.**

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2. **1–Introduction**

   - **The introduction would benefit from a more focused progression from the general waste management context to the operational problem investigated.**
   - **The motivation for improving waste collection routing is reasonable. However, please strengthen the justification for explicitly modelling stochastic intraday demand increments. Under which operating conditions does this feature materially affect routing and scheduling decisions?**
   - **The examples involving schools and restaurants illustrate possible intraday patterns. Empirical evidence or a documented practical setting would strengthen their operational relevance.**

- **The authors explain the distinction between their intraday formulation and the interday implementation of Markov et al. (2020). However, Markov et al. also discuss accommodating continuous demand information through finer time discretisation. Please clarify what substantive modelling changes the proposed formulation introduces beyond finer discretisation.**
- **The novelty discussion should address Yu, Normasari and Chen (2021), "Locationrouting problem with time-dependent demands," Computers & Industrial Engineering, 151, 106936. Although their demand accumulation is deterministic, their model already considers pickup quantities that depend on vehicle arrival time. Please explain the additional contribution of the proposed stochastic intraday formulation.**
- **The first two listed contributions appear to overlap. Introducing a new waste collection problem and incorporating stochastic intraday demand increments describe the same modelling contribution. Please consolidate these statements and distinguish the remaining contributions.**
- **The introduction explains that ML selects a policy based on instance features. Please position this contribution against existing learning-based algorithm-selection research, particularly Asín-Achá et al. (2024), "Selecting fast algorithms for the capacitated vehicle routing problem with machine learning techniques," Networks, 84, 465–480. The manuscript should explain what its candidate policies and integration with stochastic optimisation add to established portfolio-selection approaches.**

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3. **2 – Related literature**

- **Please streamline the broad waste management background and summaries of review articles. Place greater attention to the studies most relevant to the formulation and solution methodology.**
- **The review should focus primarily on stochastic and time-dependent waste collection routing, while also addressing service-time-dependent pickup models and learning-based algorithm selection.**
- **Please compare the closest studies more systematically in terms of demand evolution, service-time dependence, information availability, planning horizon and recourse decisions.**
- **Table 1 is useful, but its categories do not fully establish the claimed distinction. Please explicitly distinguish intraday from interday demand evolution and indicate whether collected quantities depend on service time.**
- **Where publication timing permits, please consider Huang et al. (2026), "Dynamic Collection Routing Optimization for Domestic Waste with Mixed Fleets," Systems, 14(5), 461. This study considers intraday waste-generation patterns and random fluctuations. Please clarify differences in demand modelling, information availability and operational decisions.**

xxx

4. **3 – Problem description and mathematical formulation**

   - **The information structure requires clarification. Demand becomes known sequentially upon arrival at collection points, whereas Section 3.2 assumes a single recourse decision moment with full knowledge of the problem data. Please explain how the two-stage representation corresponds to the operational process.**
   - **If recourse follows a predetermined rule of returning to the depot when capacity is exhausted, please demonstrate that its scenario-based implementation depends only on information available at that point. The statement that the representation is conservative also requires justification.**
   - **Section 3.3 omits the underlying stochastic programme and its second-stage conditions. Since stochastic modelling is central to the contribution, please provide a self-contained formulation or a precise explanation linking the operational problem to the scenariobased model.**
   - **Please strengthen the practical justification for the recourse restriction. Vehicles abandon the remaining route after reaching capacity, without unloading and continuing or reassigning collection points. In which operating settings is this appropriate?**
   - **The objective applies the same unit penalty to waste left because of capacity shortages and waste generated after a scheduled collection. Please justify this treatment, given that post-collection accumulation may be a normal feature of daily operations.**
   - **Constraints (7) appear to permit waiting between visits. Please clarify whether waiting and delayed departures are allowed, how they influence scheduling and whether associated costs are considered.**
   - **Please explain the convention for assigning demand increments at the beginning of each time slot and discuss sensitivity to temporal discretisation.**
   - **Please clarify how constraints (17)-(21) enforce the saturation and route-skipping rule, including the case in which cumulative demand exactly equals vehicle capacity.**
   - **Please state the assumptions concerning service times and dependence between demand increments across collection points and periods. The determination of the large constants should also be explained.**

   xxx

5. **4 – Solution methodology**

   - **Please define "policy" consistently. The candidate policies appear to be alternative deterministic optimisation procedures for constructing routing and scheduling plans.**
   - **Please justify the choice of demand approximations, capacity adjustments and adjustment percentages. A sensitivity analysis would help establish whether the results depend strongly on these choices.**

- **Please provide sufficient detail to reproduce each procedure, particularly the scheduling of visits and treatment of capacity violations in the modified VRPs.**
- **Statements suggesting that the ML framework ensures an optimal or nearly optimal policy should be qualified. Prediction does not guarantee this outcome, and the best candidate policy is not necessarily close to the stochastic optimum.**
- **Please report the ML implementation details, including hyperparameters, tuning procedures, random seeds and the tolerance used for multilabel classification.**
- **Please distinguish offline label-generation and training costs from online feature extraction, prediction and solution of the selected optimisation model.**

xxx

6. **5 – Computational results**

   - **The experiments contain only five to eight collection points. Small instances are useful for comparisons with near-optimal reference solutions, but additional experiments at larger scales are needed to support claims of practical relevance and computational efficiency.**
   - **Please make the generated instances and accompanying documentation available in a public repository. The documentation should describe the file format, parameters, units, demand-generation procedure and random seeds. Providing the instance-generation code would further support reproducibility.**
     Giuseppe
   - **Only five scenarios are used. Please justify this choice, investigate sensitivity to scenario count and evaluate the resulting plans on independent demand realisations.**
   - **Please document the instance-generation procedure more fully, including collection point type assignment, travel costs and times, penalty coefficients, demand dependence and random seeds.**
   - **Synthetic instances can support a methodological study, but their demand patterns and parameter ranges should be justified in relation to relevant operating conditions.**
   - **To isolate the benefit of the modelling contribution, please compare the proposed approach with a comparable formulation that omits intraday demand variation. The stochastic-versus-expected-value comparison alone does not establish the value of intraday modelling.**
   - **Please distinguish instances for which no feasible solution was found within the time limit from those proven infeasible. The recurring proportion of "not solved" instances in Table 5 needs explanation.**
   - **The VSS-type measure uses computed solution values, some associated with substantial optimality gaps. Please distinguish estimates from certified values and discuss how incomplete optimisation affects their interpretation.**

- **The policy and ML experiments use 178 instances selected for near-optimal solvability and a VSS of at least 4%. Please assess performance on a broader test population or explicitly limit the conclusions to this selected subset.**
- **The ML evaluation uses one split containing 115 training and 63 test instances. Please strengthen the assessment through repeated splits or cross-validation and report variability in performance. Any tuning should remain independent of the final evaluation data.**
  Alessio
- **Please clarify whether P7 was selected as the best fixed policy using training data only. Baseline selection should be independent of test-set outcomes.**
- **The Ideal-policy selects the best of the twelve candidate procedures. It is a bestachievable reference within that portfolio, rather than a lower bound on the original stochastic optimisation problem. A 3% gap to this reference therefore does not demonstrate near-optimality for the WCRP-TVSD.**
- **Please compare the ML-selected solutions directly with reliable stochastic-model reference values where available, alongside the comparison with the Ideal-policy.**
- **Table 7 should include end-to-end computational times for the ML approaches and relevant baselines. Since the candidate policies have substantially different runtimes, executing one policy does not by itself establish the claimed computational advantage.**
- **Please interpret training and test results separately. The combined OVERALL results do not provide evidence of generalisation, and claims of reliable performance should reflect the limited evaluation.**

xxx

7. **6 – Conclusions**

- **Please distinguish the specific modelling contribution from established elements such as stochastic programming and learning-based policy selection.**
- **Claims of practical effectiveness, reliability and near-optimality should be moderated to match the evidence, or supported by the additional experiments requested above.**
- **Please identify the reported 3% gap explicitly as a gap to the best candidate policy, rather than to the stochastic optimum.**
- **Please state the experimental limitations prominently, including the small synthetic instances, limited scenario count and selected ML dataset.**
- **The conclusion describes the ML selector in terms of classification algorithms, although the best-performing approach uses regression. Please correct this description.**
- **Please clarify that the average VSS of 21.33% is calculated over the 597 instances with reported feasible solutions, rather than all 720 generated instances.**

xxx

I appreciate the authors' efforts and the potential value of the research question. A clearer positioning of the contribution, a rigorous explanation of the information structure and substantially stronger computational validation are necessary before the manuscript can be considered for publication.
Disclose statement: During the preparation of these comments, I used ChatGPT to help refine and organise my comments and identify relevant literature. After using this tool, I reviewed and edited the content as needed, and I take full responsibility for the report's content.